AP BIOLOGY TOPIC 8.1

Responses to the Environment: Behavior, Tropisms & Communication

UNIT 8: ECOLOGY (10–15% EXAM WEIGHT)

ENE-3.D | IST-5.A | TAXIS VS. KINESIS | PHOTOTROPISM & AUXIN

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§1. KEY TERMS & FORMAL DEFINITIONS LEXICON — TOPIC 8.1 (RESPONSES TO THE ENVIRONMENT: BEHAVIOR, TROPISMS & COMMUNICATION)

CED Core Vocabulary

Behavioral Responses (External Actions) — Rapid movement or activity changes in response to stimuli — moving to favorable microclimates, avoiding predators, foraging, migrating, hibernating, or...

Physiological Responses (Internal Adjustments) — Biochemical, hormonal, or systemic changes — thermoregulation (sweating, shivering, vasodilation/vasoconstriction), osmoregulation, circadian rhythms, and...

Innate vs. Learned Behaviors — Innate Behavior: Genetically programmed, inherited, automatic ("hard-wired") behaviors performed without prior experience (e.g., Fixed Action Patterns [FAPs]...

Learned Behavior — Modified through experience and environmental interaction; flexible (e.g., imprinting in young geese [Konrad Lorenz], spatial learning, classical/operant...

1. Chemical (Pheromones) — Volatile or water-borne molecules used to mark trails (ants), signal reproductive readiness (female silkworm moth bombykol), or trigger alarm (minnows)....

2. Auditory (Sound) — Bird songs (attract mates + defend territory), whale songs, frog calls, vervet monkey predator-specific alarm calls.

3. Visual — Courtship displays (peacock train, firefly bioluminescence), threat postures, and aposematic warning coloration (bright red/yellow/black in poison dart...

4. Tactile (Touch) — Primate social grooming, Honeybee Waggle Dance (Karl von Frisch) inside dark hives — angle relative to vertical communicates food direction relative to the sun;...

§2. 1. BEHAVIORAL & PHYSIOLOGICAL RESPONSES TO CUES

Homeostasis & Fitness

HIERARCHICAL CONCEPT & PATHWAY MAP **[CAUSE ➔ MECHANISM ➔ OUTCOME]**

1. Behavioral & Physiological Responses to Cues ➔

Behavioral Responses (External Actions) ➔ **Innate vs. Learned Behaviors**

- Organisms respond to **internal and external environmental stimuli** through coordinated **behavioral and physiological mechanisms** that optimize survival, reproduction (evolutionary fitness), and homeostasis.
- **Behavioral Responses (External Actions):** Rapid movement or activity changes in response to stimuli — moving to favorable microclimates, avoiding predators, foraging, migrating, hibernating, or courtship.
  - *Example:* Desert lizards bask on sunlit rocks when cold and retreat into cool burrows at midday.
- **Physiological Responses (Internal Adjustments):** Biochemical, hormonal, or systemic changes — thermoregulation (sweating, shivering, vasodilation/vasoconstriction), osmoregulation, circadian rhythms, and **acclimatization** (e.g., producing more red blood cells via EPO at high altitude).
- **Innate vs. Learned Behaviors:**
  - **Innate Behavior:** Genetically programmed, inherited, automatic ("hard-wired") behaviors performed without prior experience (e.g., **Fixed Action Patterns [FAPs]** like stickleback fish attacking red bellies or graylag goose egg-rolling; infant suckling reflex).
  - **Learned Behavior:** Modified through experience and environmental interaction; flexible (e.g., **imprinting** in young geese [Konrad Lorenz], spatial learning, classical/operant conditioning, social learning).

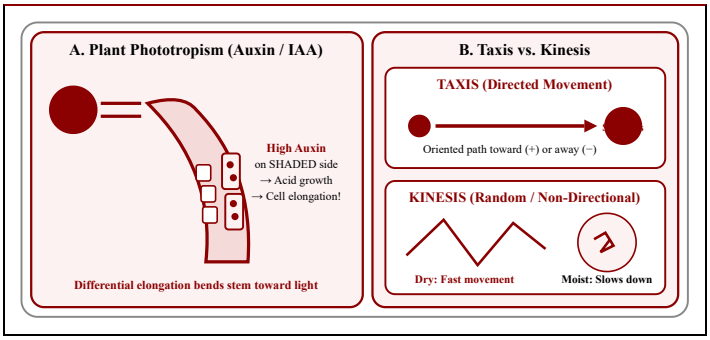
[EXAMPLE & BIOLOGICAL APPLICATION] **Behavior + Physiology Work**

Together: A human in freezing weather responds **behaviorally** by putting on a winter coat/huddling and **physiologically** by shivering and vasoconstricting peripheral skin arterioles.

[COMMON AP EXAM FRQ TRAP] **Acclimatization is NOT a behavioral response and NOT evolution!** Acclimatization is a reversible **physiological adjustment** within an individual's lifetime (e.g., growing a thicker winter coat or increasing RBC count at high altitude).

§4. VISUALIZING PLANT PHOTOTROPISM & TAXIS VS. KINESIS

Experimental Models



§5. 3. ANIMAL COMMUNICATION, COOPERATION & KIN SELECTION

IST-5.A Signals

- **5 Signal Modalities Used to Exchange Information & Alter Behavior:**
  - **1. Chemical (Pheromones):** Volatile or water-borne molecules used to mark trails (ants), signal reproductive readiness (female silkworm moth *bombykol*), or trigger alarm (minnows). Effective day/night and over long durations.
  - **2. Auditory (Sound):** Bird songs (attract mates + defend territory), whale songs, frog calls, vervet monkey predator-specific alarm calls.
  - **3. Visual:** Courtship displays (peacock train, firefly bioluminescence), threat postures, and **aposematic warning coloration** (bright red/yellow/black in poison dart frogs/monarchs).
  - **4. Tactile (Touch):** Primate social grooming, **Honeybee Waggle Dance (Karl von Frisch)** inside dark hives — angle relative to vertical communicates food *direction* relative to the sun; waggle duration communicates *distance*!
  - **5. Electrical:** Electroreception and signaling in aquatic sharks, rays, and electric knifefish.
- **Cooperative Behavior & Altruism (Kin Selection / Inclusive Fitness):**
  - **Pack hunting** (wolves, lions), **schooling/flocking**, and **eusociality** (bees, ants, naked mole-rats) increase group survival and foraging efficiency.
  - **Altruism & Hamilton's Rule (rB > C):** Self-sacrificing behavior (e.g., Belding's ground squirrel giving an alarm call that attracts predators, or sterile worker bees raising the queen's offspring) is favored by natural selection when the benefit (**B**) to close genetic relatives weighted by the **coefficient of relatedness** (**r**) exceeds the reproductive cost (**C**) to the altruist — increasing **inclusive fitness**!

§3. 2. ANIMAL MOVEMENT (TAXIS VS. KINESIS) & PLANT RESPONSES

Directed vs. Random

RESPONSE TYPE	MECHANISM & DIRECTIONALITY	HIGH-YIELD AP EXAMPLES
Taxis	**Directed, oriented movement** directly toward (*positive taxis*) or away from (*negative taxis*) a stimulus.	• **Positive Phototaxis:** Moths / photosynthetic *Euglena* moving toward light. • **Chemotaxis:** Bacterial cells swimming up a glucose concentration gradient; trout swimming upstream (*rheotaxis*).
Kinesis	**Random, undirected change in activity/turning rate** in response to stimulus intensity (not oriented toward/away).	• **Sowbugs / Pillbugs (Isopods):** Move fast and straight in dry, bright areas (*high kinesis*), but slow down and turn frequently in moist, dark areas → end up aggregating in damp habitats!
Plant Phototropism	Directional growth of plant shoots toward light driven by **Auxin (** **IAA** **)** hormone redistribution.	Auxin migrates from the lit side to the **shaded side of the coleoptile tip** → stimulates proton pumps (**H^+**) that loosen cellulose walls → **shaded cells elongate faster**, bending shoot toward light!
Photoperiodism	Plant physiological response (flowering/dormancy) to **night length (darkness duration)** via **phytochrome** photoreceptors (**$P_r \leftrightarrow P_{fr}$**).	**Short-day (long-night) plants** flower only when uninterrupted darkness exceeds a critical threshold; a brief flash of red light (**660 nm**) interrupts night and blocks flowering (reversed by far-red **730 nm**).

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§6. SCHOLARLY SYNTHESIS: EXPERIMENTAL DESIGN, QUANTITATIVE ANALYSIS, PATHOLOGY & AP FRQ RHETORIC

High-Yield Exam Rigor

MASTER TOPIC SYNTHESIS & CAUSAL ARCHITECTURE — TOPIC 8.1 **[MOLECULAR CAUSE ➔ CELLULAR MECHANISM ➔ PHYSIOLOGICAL / ECOLOGICAL EFFECT ➔ FRQ JUSTIFICATION]**

Topic 8.1: Responses to the Environment: Behavior, Tropisms & Communication ➔ **Structural / Energetic Determinants** ➔ **Regulated Pathway & Homeostasis** ➔ **Experimental Perturbation (±2 $SE_{\bar{x}}$)** ➔ **Phenotypic / Evolutionary Outcome**

SCHOLARLY DIMENSION	COMPREHENSIVE ACADEMIC REFERENCE & MECHANISTIC SPECIFICATION
I. Experimental Design & Required Lab Controls	AP Investigations 10, 11 & 12 (Energy Dynamics, Whole-Plant Transpiration & Pillbug Animal Behavior Choice Chambers): Choice chambers testing pillbug (Armadillidium vulgare) chemotaxis/hydrotaxis/kinesis (count pillbugs on wet vs. dry filter paper every 30 s for 10 min; evaluate using χ^2 against 50:50 null distribution). Potometer/gravimetric transpiration across light, wind, and humidity treatments normalized per cm^2 leaf surface area. **Null Hypothesis (H_0):** State explicitly that the independent variable treatment has *no statistically significant effect* on the dependent variable; any observed difference is due to random sampling chance alone.
II. Quantitative Formulas & Graph Interpretation	Quantitative Formulas & Ecological Equations: Population Growth Rate: $dN/dt = B - D$. Exponential Growth: $dN/dt = r_{max} \cdot N$ (J-curve). Logistic Growth: $dN/dt = r_{max} \cdot N \cdot [(K - N)/K]$ (S-curve; maximum instantaneous growth rate dN/dt occurs at $N = K/2$). Simpson's Diversity Index: $D_I = 1 - \Sigma(n / N)^2$ (ranges from 0 to 1; higher value = higher diversity). Net Primary Productivity: $NPP = GPP - R$. Trophic 10% Rule: ~10% of energy transfers to the next trophic level (~90% lost as metabolic heat/respiration). **Error Bar Rule (±2 $SE_{\bar{x}}$):** If the ±2 $SE_{\bar{x}}$ error bars of two treatment groups *do not overlap*, there is a statistically significant difference between means ($p < 0.05$; reject H_0).
III. Pathology, Mutation & Disruption Case Study	Ecological Disruption & Trophic Cascade Case Study: Removal of a keystone predator (e.g., Pisaster ochraceus sea star in rocky intertidal zones, or sea otters in kelp forests) triggers a top-down trophic cascade: prey populations (mussels or sea urchins) explode unchecked, competitively excluding other invertebrates or overgrazing kelp producers, collapsing Simpson's Diversity Index. Similarly, agricultural N/P fertilizer runoff causes Cultural Eutrophication: algal blooms block light, aerobic bacteria decompose dead algae and deplete dissolved O_2, creating hypoxic dead zones.
IV. AP FRQ Task Verb Scoring Templates	• **DESCRIBE / IDENTIFY:** State the direct trend, value, or structural component with exact units from the figure. • **EXPLAIN (How/Why):** Connect cause to effect via a 3-link chain: *"[Change in X] causes [molecular/cellular mechanism Y], which results in [observable outcome Z]."* • **PREDICT:** State the specific directional change (increase, decrease, or remain constant) when the pathway is perturbed. • **JUSTIFY:** Pair quantitative data/evidence from the prompt with the underlying biological principle from Topic 8.1 to prove why the claim holds.